

Kinetic Modeling of Cell Signaling

Module 1: Reaction networks as dynamical systems

Jim Faeder

MSCBIO 2025 · Introduction to Bioinformatics Programming in Python

Fall 2026

Resources

- ▶ **BioNetGen** software, documentation, and tutorials:
bionetgen.org
- ▶ **VS Code extension:** search the Extensions panel for *BioNetGen*
- ▶ **BNG Playground:** write and simulate models in your browser, no installation
ruleworld.github.io/bngplayground
- ▶ **bngsim:** fast ODE and stochastic simulation from Python
bngsim.readthedocs.io · `pip install bngsim bionetgen`
- ▶ **Course repository:** models, notebooks, slides and handouts
github.com/jrfaeder/RBM_Intro_BioNetGen



Course repository



BNG Playground

Additional reading

- ▶ J. R. Faeder, M. L. Blinov, and W. S. Hlavacek (2009). **Rule-based modeling of biochemical systems with BioNetGen.** *Methods Mol. Biol.* **500**, 113–167. [doi:10.1007/978-1-59745-525-1_5](https://doi.org/10.1007/978-1-59745-525-1_5)
A thorough introduction to BNGL and the BioNetGen software.
- ▶ J. A. P. Sekar and J. R. Faeder (2012). **Rule-based modeling of signal transduction: a primer.** *Methods Mol. Biol.* **880**, 139–218. [doi:10.1007/978-1-61779-833-7_9](https://doi.org/10.1007/978-1-61779-833-7_9)
A self-contained tutorial: building a receptor model step by step, model assumptions, compartments, and a library of example rules.
- ▶ L. A. Harris et al. (2016). **BioNetGen 2.2: advances in rule-based modeling.** *Bioinformatics* **32**, 3366–3368. [Free full text \(PMC\)](#)
What version 2.2 added: rate laws beyond mass action, faster stochastic and hybrid simulation, and SBML import.

The two *Methods in Molecular Biology* chapters are available through the Pitt library: follow the DOI link on campus or after signing in to the library.

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4. encode the same network in the **BioNetGen language** (BNGL), simulate it, and interpret the results
5. explain how a **negative feedback** loop shapes the dynamics of a signaling response

Part 1

Why model reaction networks?

Reaction networks describe how molecule populations change

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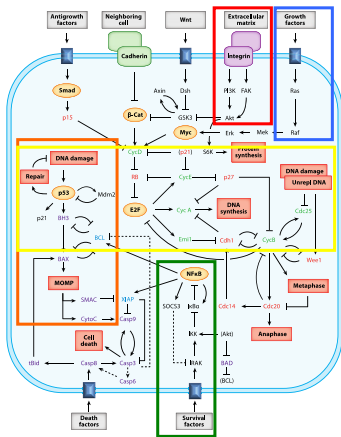
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What a model gives us

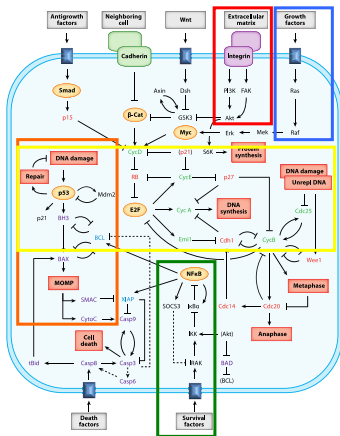
- ▶ A precise statement of a mechanism
- ▶ Predictions of dynamics we cannot easily measure
- ▶ A way to test whether a mechanism can explain data
- ▶ Hypotheses for the next experiment

Toward an integrated model of the cell



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Toward an integrated model of the cell

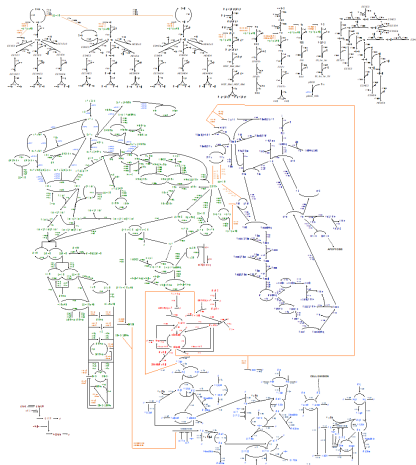
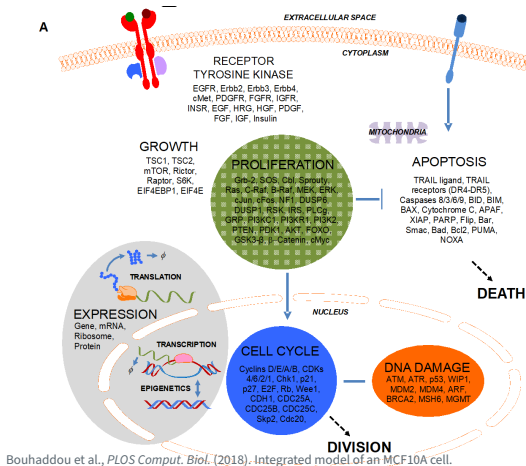


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How do we link these modules together into models that predict how cells respond?

Tyson and Novak, *Annu. Rev. Phys. Chem.* **61**, 219–240 (2010)

Large models are built from simple reaction steps

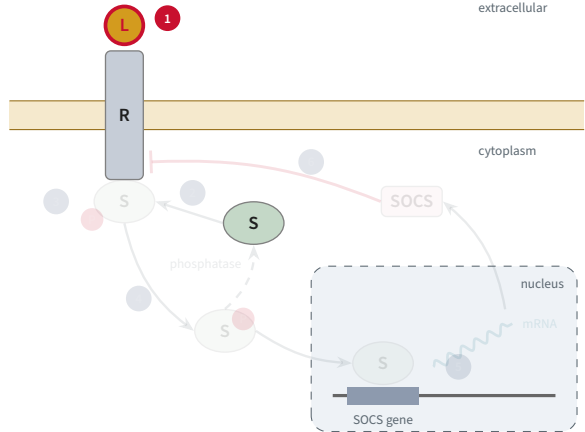


Part 2

A cytokine signaling circuit

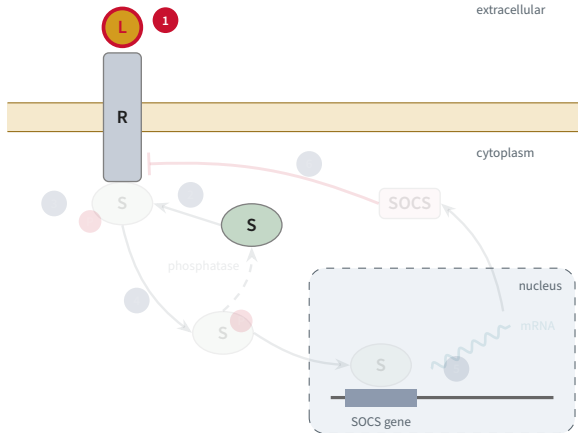
Cytokines signal through receptors, STATs, and SOCS feedback

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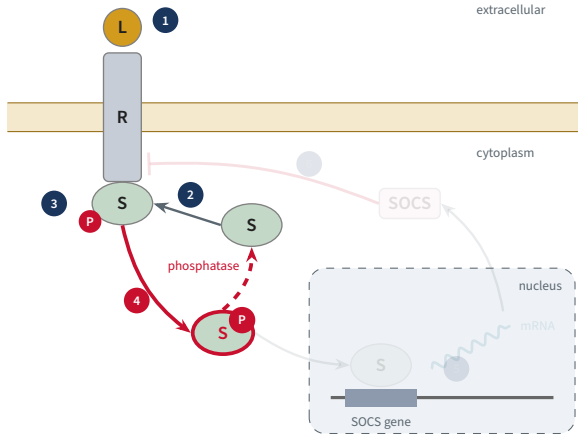
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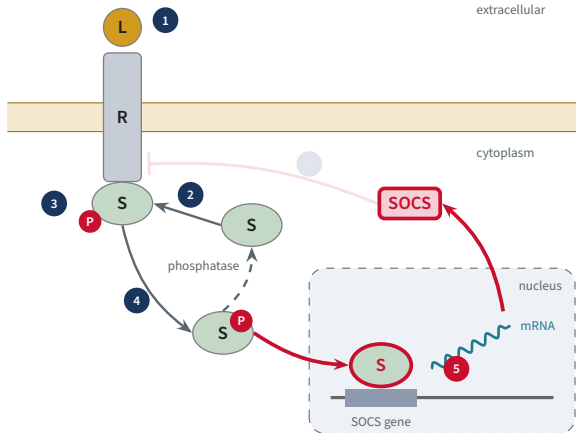
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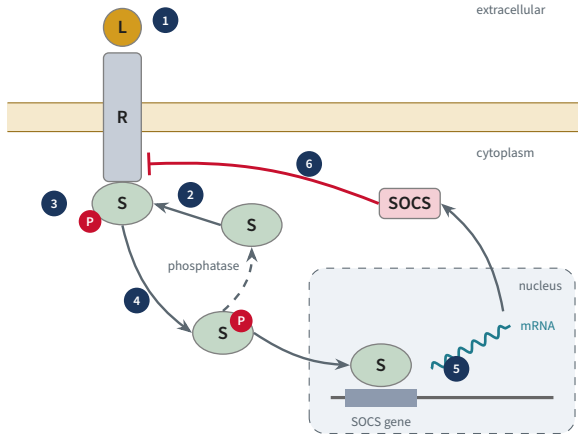
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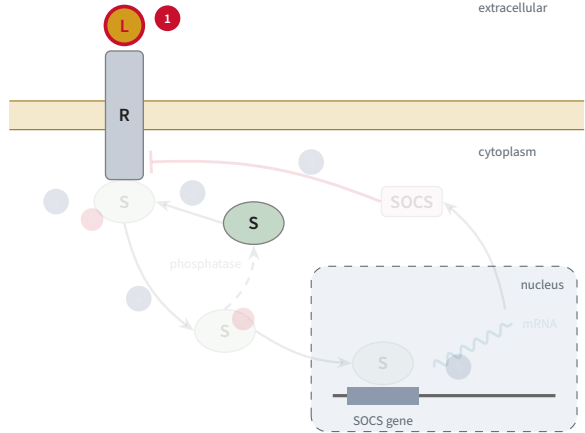
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- ▶ pSTAT induces genes including **SOCS** (suppressor of cytokine signaling).
- ▶ SOCS binds the receptor and **shuts off further STAT activation**: negative feedback.



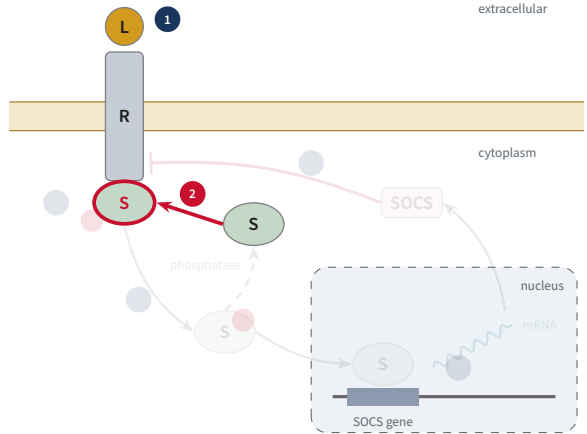
We will build the model one reaction step at a time

1. Ligand-receptor binding



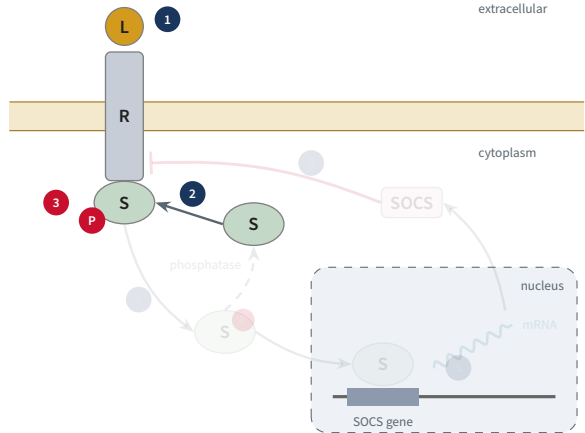
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2. STAT binds the ligand-bound receptor



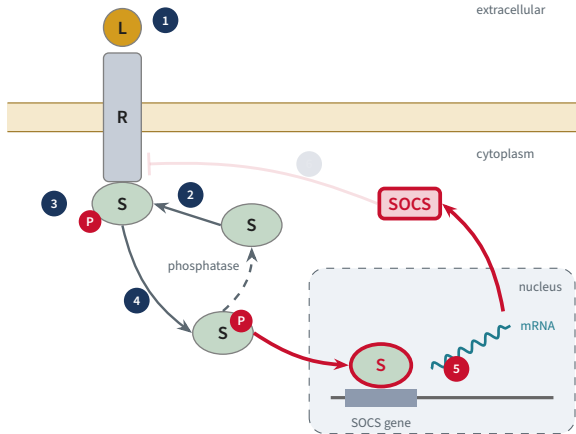
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3. Receptor-bound STAT is activated



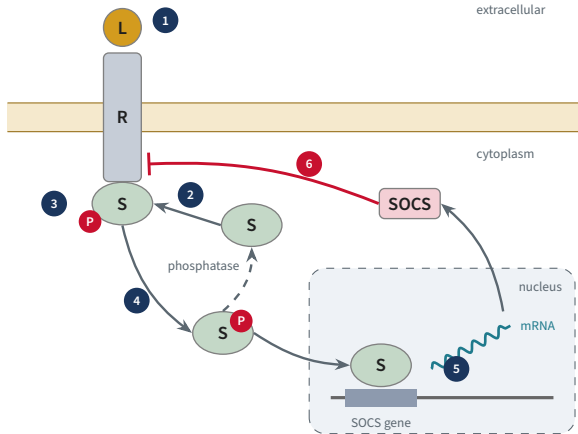
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1. Ligand–receptor binding
2. STAT binds the ligand-bound receptor
3. Receptor-bound STAT is activated
4. Activated STAT dissociates
5. pSTAT binds the SOCS promoter; SOCS is transcribed and translated



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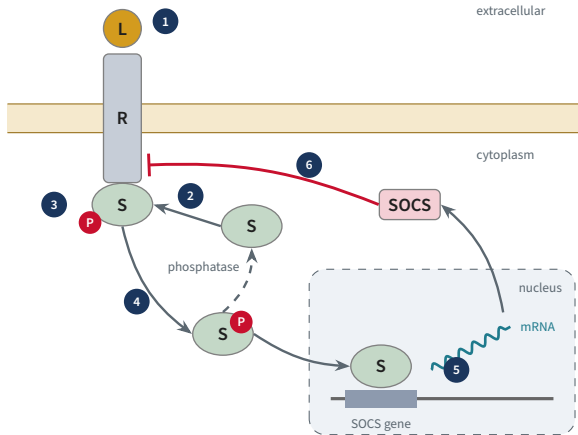
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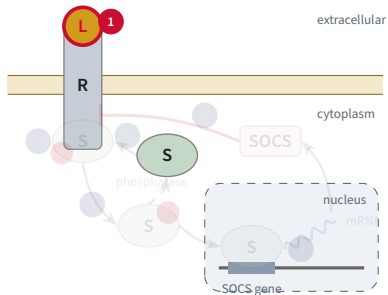
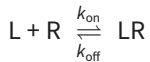
Each step adds reactions to the previous model.
The files are `models/stat_step1_binding.bngl`
through `stat_step6_feedback.bngl`.



Part 3

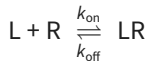
Step 1: Ligand–receptor binding

Binding is a reversible pair of reactions



Units: k_{on} in $\mu\text{M}^{-1}\text{min}^{-1}$, k_{off} in min^{-1}

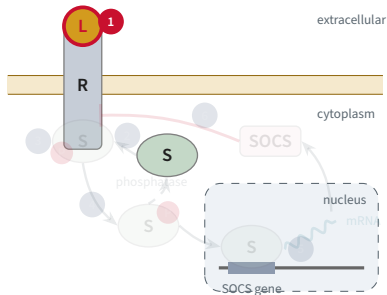
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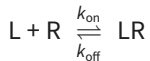
binding: $v_1 = k_{\text{on}} [L] [R]$

unbinding: $v_2 = k_{\text{off}} [LR]$



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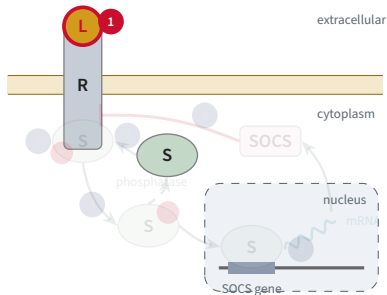
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Law of mass action

The rate of an elementary reaction is proportional to the product of the concentrations of its reactants. The proportionality constant is the **rate constant**.



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Each reaction contributes to the rate of change of its species

	v_1 (bind)	v_2 (unbind)
L	-1	+1
R	-1	+1
LR	+1	-1

Reactants are consumed ($-$), products are created ($+$).

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Ordinary differential equations (ODEs) follow mechanically from the list of reactions and their rate laws.

The stoichiometry matrix gives a compact general form

Collect the concentrations in a vector $\mathbf{X}$ and the reaction rates in $\mathbf{v}$:

$$\frac{d\mathbf{X}}{dt} = \mathbf{S} \mathbf{v}(\mathbf{X}), \quad \frac{d}{dt} \begin{pmatrix} [L] \\ [R] \\ [LR] \end{pmatrix} = \underbrace{\begin{pmatrix} -1 & +1 \\ -1 & +1 \\ +1 & -1 \end{pmatrix}}_{\mathbf{S}} \underbrace{\begin{pmatrix} k_{\text{on}}[L][R] \\ k_{\text{off}}[LR] \end{pmatrix}}_{\mathbf{v}(\mathbf{X})}$$

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- ▶ $\mathbf{S}$ has one row per species and one column per reaction.
- ▶ This form holds for *any* reaction network; only $\mathbf{S}$ and $\mathbf{v}$ change.

Conservation laws and equilibrium

Conservation. Binding only moves molecules between free and bound forms:

$$[L] + [LR] = L_0$$

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Small models can be solved by hand. Real signaling networks cannot, so we solve the ODEs numerically.

ODEs for reaction networks are solved numerically

Simplest idea (forward Euler): take small time steps

Δt ,

$$\mathbf{X}(t + \Delta t) \approx \mathbf{X}(t) + \Delta t \mathbf{S} \mathbf{v}(\mathbf{X}(t))$$

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In practice:

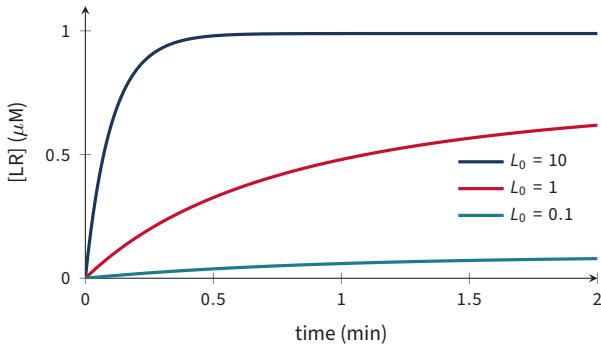
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In BioNetGen

```
simulate({method=>"ode", ...})  
uses CVODE.
```

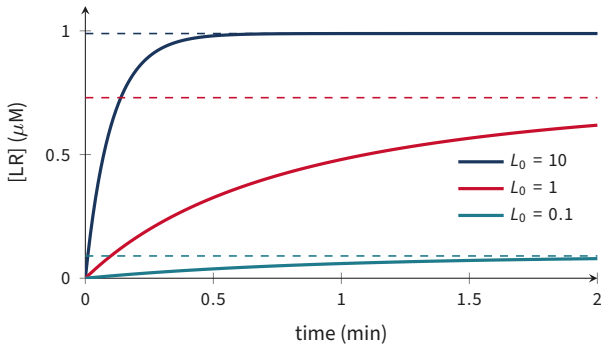
In Python, `bngsim.Simulator(model, method="ode")` runs the same model with CVODE.

Binding approaches equilibrium faster and further at higher ligand



$$R_0 = 1 \mu\text{M}, k_{\text{on}} = 1 \mu\text{M}^{-1}\text{min}^{-1}, \\ k_{\text{off}} = 0.1 \text{ min}^{-1} (K_D = 0.1 \mu\text{M}).$$

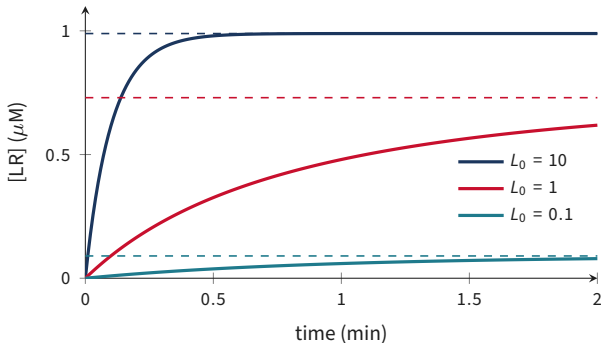
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Dashed lines: $[LR]_{\text{eq}}$ from the formula on the previous slide.

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The relaxation rate is $k_{\text{on}}([L] + [R]) + k_{\text{off}}$ near equilibrium, so higher ligand also means a faster response.

The same model in BNGL

```
begin model

begin parameters
  L0      10      # ligand (uM)
  R0      1       # receptor (uM)
  kon     1       # binding (1/uM/min)
  koff    0.1     # unbinding (1/min)
end parameters

begin molecule types
  L
  R
  LR
end molecule types

begin species
  L    L0
  R    R0
end species
```

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end species
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```
begin observables
  Molecules Lfree  L
  Molecules Rfree  R
  Molecules LR     LR
end observables

begin reaction rules
  # Ligand binds receptor
  LRbind: L + R <=> LR kon, koff
end reaction rules

end model

generate_network({overwrite=>1})
simulate({method=>"ode", t_end=>2, n_steps=>200})
```

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  Molecules Lfree  L
  Molecules Rfree  R
  Molecules LR     LR
end observables

begin reaction rules
  # Ligand binds receptor
  LRbind: L + R <=> LR kon, koff
end reaction rules

end model

generate_network({overwrite=>1})
simulate({method=>"ode", t_end=>2, n_steps=>200})
```

► **observables:** quantities to output

The same model in BNGL

```
begin model

begin parameters
  L0      10    # ligand (uM)
  R0      1     # receptor (uM)
  kon     1     # binding (1/uM/min)
  koff    0.1   # unbinding (1/min)
end parameters

begin molecule types
  L
  R
  LR
end molecule types

begin species
  L    L0
  R    R0
end species
```

```
begin observables
  Molecules Lfree  L
  Molecules Rfree  R
  Molecules LR     LR
end observables

begin reaction rules
  # Ligand binds receptor
  LRbind: L + R <-> LR kon, koff
end reaction rules

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generate_network({overwrite=>1})
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- ▶ **observables:** quantities to output
- ▶ **reaction rules:** $\leftrightarrow$ is reversible; forward and reverse rate constants

The same model in BNGL

```
begin model

begin parameters
  L0      10      # ligand (uM)
  R0      1       # receptor (uM)
  kon     1       # binding (1/uM/min)
  koff    0.1     # unbinding (1/min)
end parameters

begin molecule types
  L
  R
  LR
end molecule types

begin species
  L    L0
  R    R0
end species
```

```
begin observables
  Molecules Lfree  L
  Molecules Rfree  R
  Molecules LR     LR
end observables

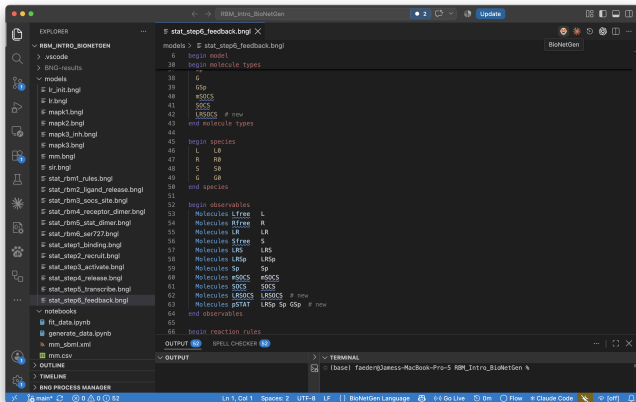
begin reaction rules
  # Ligand binds receptor
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end reaction rules

end model

generate_network({overwrite=>1})
simulate({method=>"ode", t_end=>2, n_steps=>200})
```

- ▶ **observables:** quantities to output
- ▶ **reaction rules:** $\leftrightarrow$ is reversible; forward and reverse rate constants
- ▶ **actions** after `end model`: generate the network, then simulate

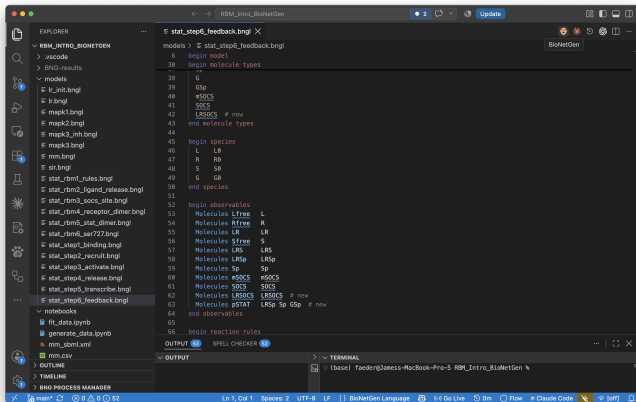
Running a model in VS Code



► Open a .bngl file

VS Code extension: Ali Sinan Saglam

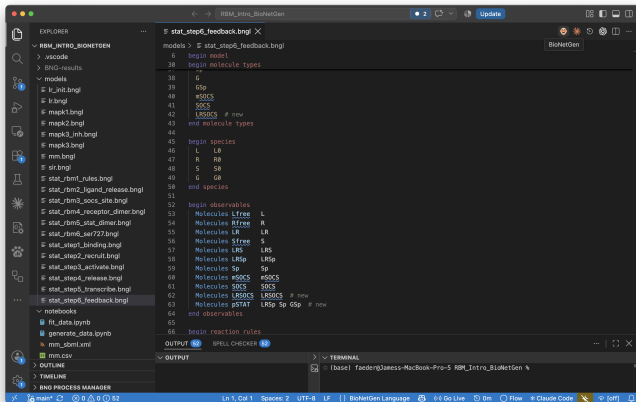
Running a model in VS Code



- ▶ Open a .bngl file
- ▶ Click the **run** button in the editor toolbar

VS Code extension: Ali Sinan Saglam

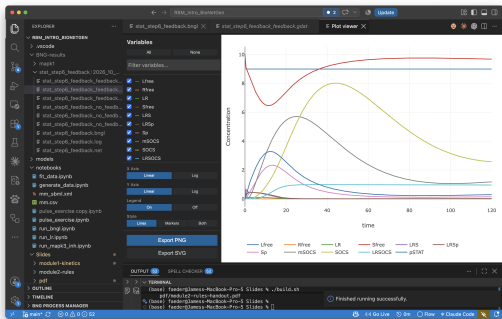
Running a model in VS Code



- ▶ Open a .bngl file
- ▶ Click the **run** button in the editor toolbar
- ▶ Results go to BNG-results/<model>/<timestamp>/

VS Code extension: Ali Sinan Saglam

Interactive plotting and Python notebooks



Built-in plotting of .gdat output

BNG Playground: BioNetGen in your browser

The screenshot displays the BNG Playground web interface. On the left is the BNGL Editor, which contains a text area with BNGL code for a binding model. The code defines parameters for initial molecule counts (A0, B0), association/dissociation rate constants (ka, kd), and molecule types (A, B, AB). It also sets seed species, observables, and reaction rules. On the right is the Regulatory Graph, which visualizes the model as a network. It shows two input nodes, B(a) and A(b), both pointing to a central node labeled _R1. This node then points to an output node labeled AB(1),B(a(1)). The interface includes tabs for Time Courses, Network, Analysis, Regulatory, Contact Map, Rules, Influence, and Analysis. A legend at the bottom right identifies the symbols used in the graph: Rule (purple circle), AtomicPattern (orange rectangle), Interaction (black arrow), and Context (grey rectangle).

```
1 begin model
2   begin parameters
3     A0 100 # Initial number of A molecules
4     B0 100 # Initial number of B molecules
5     ka 0.1 # A-B association rate constant (1/molecule 1/s)
6     kd 1 # A-B dissociation rate constant (1/s)
7   end parameters
8   begin molecule types
9     A(a)
10    B(b)
11  end molecule types
12  begin seed species
13    A(a) A0
14    B(b) B0
15  end seed species
16  begin observables
17    Molecules A A(a)
18    Molecules B B(b)
19    Molecules C A(b(1),B(a(1))
20  end observables
21  begin reaction rules
22    A(a) + B(b) <=> A(b(1),B(a(1)) ka, kd
23  end reaction rules
24 end model
25
26 simulate(method="cde",t_end=10,n_steps=200)
27
```

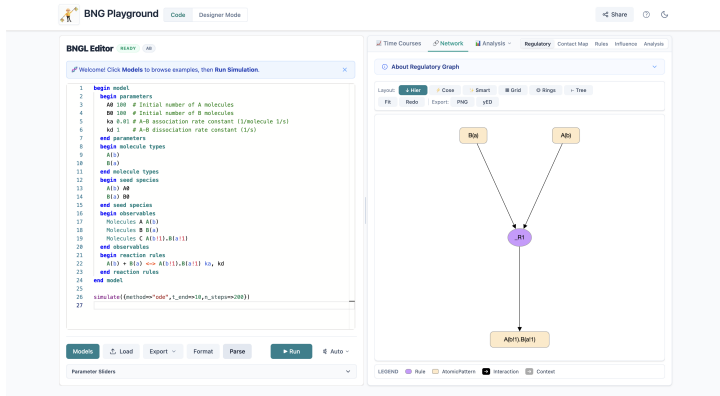
BNGL editor (left) and the regulatory graph of a binding rule (right)

- No installation: write, simulate, and analyze models



ruleworld.github.io/bngplayground

BNG Playground: BioNetGen in your browser



BNGL editor (left) and the regulatory graph of a binding rule (right)

- ▶ No installation: write, simulate, and analyze models
- ▶ Built-in examples



ruleworld.github.io/bngplayground

BNG Playground: BioNetGen in your browser

The screenshot displays the BNG Playground web interface. On the left is the BNGL Editor, which contains a text area with a BNGL model definition. The model includes parameters for initial numbers of molecules (A0, B0), association and dissociation rate constants (ka, kd), and a reaction rule for the binding of A and B to form a complex AB. The right side of the interface shows the 'About Regulatory Graph' panel, which visualizes the reaction rule as a directed graph. The graph has three nodes: 'B(a)' and 'A(b)' at the top, both pointing to a central node 'AB(1).B(a)(1)'. Below this node is another node 'AB(1).B(a)(1)'. The interface also includes a 'Run' button and a 'Parameter Sliders' section at the bottom left.

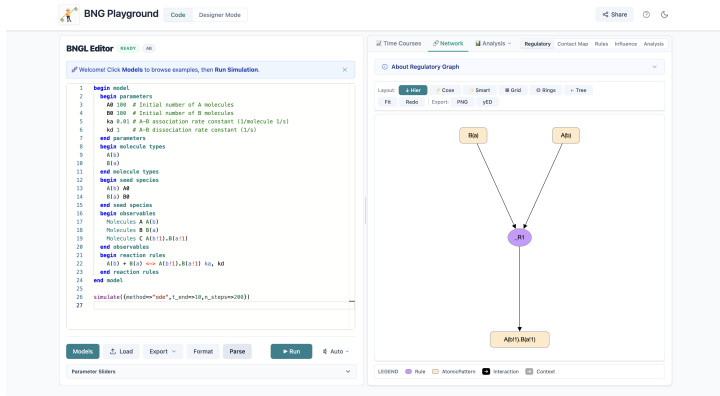
BNGL editor (left) and the regulatory graph of a binding rule (right)

- ▶ No installation: write, simulate, and analyze models
- ▶ Built-in examples
- ▶ Share a model as a link



ruleworld.github.io/bngplayground

BNG Playground: BioNetGen in your browser



BNGL editor (left) and the regulatory graph of a binding rule (right)

- ▶ No installation: write, simulate, and analyze models
- ▶ Built-in examples
- ▶ Share a model as a link
- ▶ Paste in any file from models/

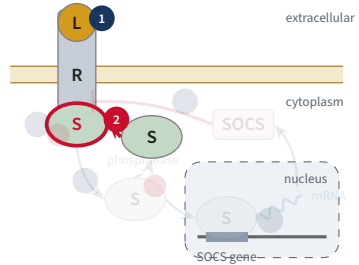
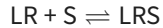


ruleworld.github.io/bngplayground

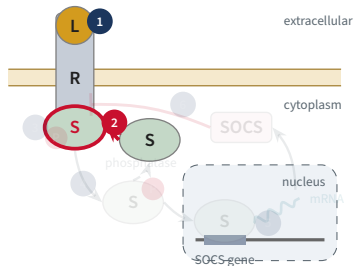
Part 4

Building the signaling circuit in BNGL

Step 2: STAT binds the ligand-bound receptor



Step 2: STAT binds the ligand-bound receptor



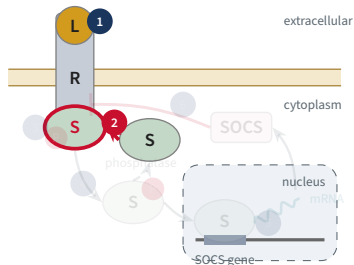
STAT can bind *only* the ligand-bound receptor LR, never free R. This is how ligand binding **enables** STAT recruitment.

Step 2: STAT binds the ligand-bound receptor



New lines in `stat_step2_recruit.bngl`:

```
begin parameters
  S0      10    # STAT (uM)
  kSon    1     # STAT binding (1/uM/min)
  kSoff   1     # STAT unbinding (1/min)
end parameters
begin molecule types
  S
  LRS
end molecule types
begin species
  S S0
end species
begin reaction rules
  Sbind: LR + S <-> LRS kSon, kSoff
end reaction rules
```



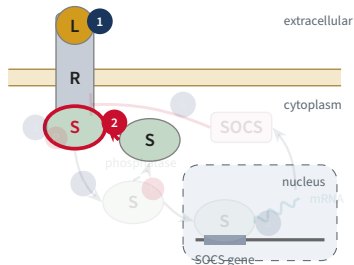
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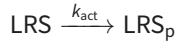
```
begin parameters
  S0      10    # STAT (uM)
  kSon    1     # STAT binding (1/uM/min)
  kSoff   1     # STAT unbinding (1/min)
end parameters
begin molecule types
  S
  LRS
end molecule types
begin species
  S S0
end species
begin reaction rules
  Sbind: LR + S <-> LRS kSon, kSoff
end reaction rules
```



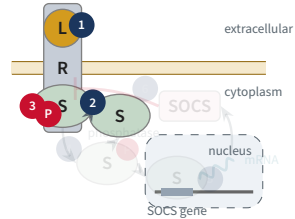
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Every new species needs a molecule type; species present at $t = 0$ also need an initial amount.

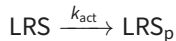
Step 3: receptor-bound STAT is activated



A receptor-associated kinase (JAK) phosphorylates the docked STAT: a **first-order**, irreversible reaction with $v = k_{\text{act}}[\text{LRS}]$.



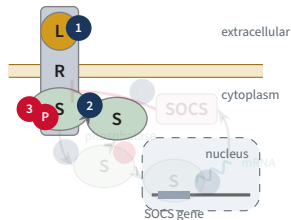
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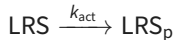
A receptor-associated kinase (JAK) phosphorylates the docked STAT: a **first-order**, irreversible reaction with $v = k_{\text{act}}[\text{LRS}]$.

```
kact 1    # STAT activation (1/min)
LRSp
Sact: LRS -> LRSp kact
```

one new line each in parameters, molecule types, and reaction rules



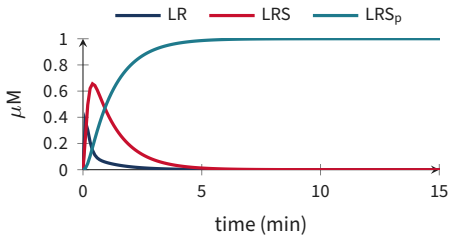
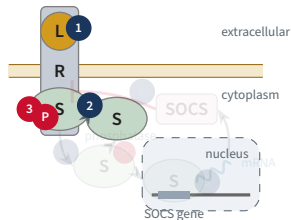
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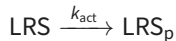
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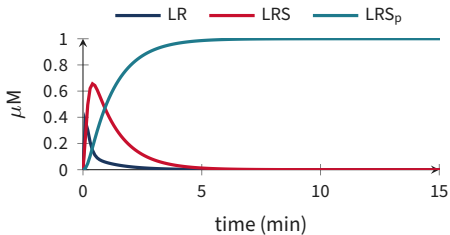
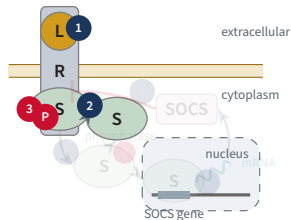


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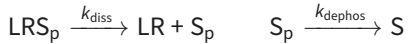
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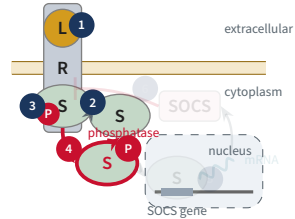
Problem: pSTAT never leaves the receptor. Every receptor ends up stuck as LRS_p .



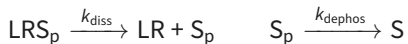
Step 4: activated STAT dissociates from the receptor



Releasing pSTAT frees the receptor to activate another STAT. A phosphatase returns free pSTAT to the inactive pool.

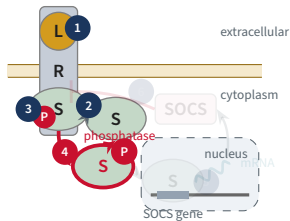


Step 4: activated STAT dissociates from the receptor

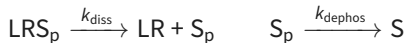


Releasing pSTAT frees the receptor to activate another STAT. A phosphatase returns free pSTAT to the inactive pool.

```
kdiss    1    # pSTAT release (1/min)
kdephos  0.1  # dephosphorylation (1/min)
Sp
Srelease: LRSp -> LR + Sp kdiss
Sdephos:  Sp -> S kdephos
```

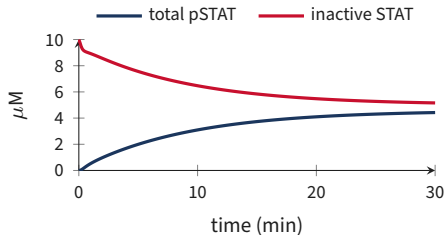
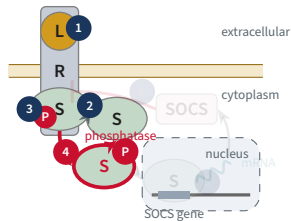


Step 4: activated STAT dissociates from the receptor

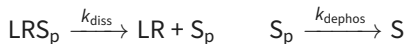


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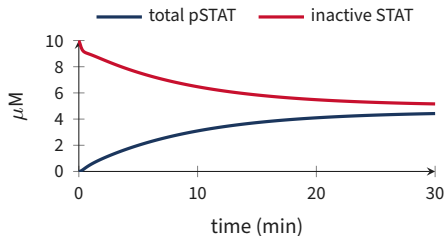
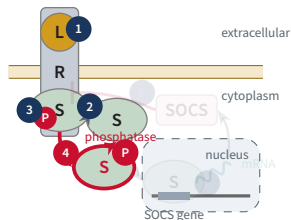
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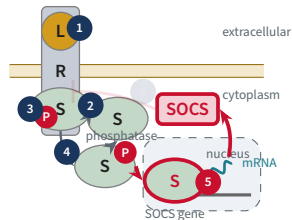
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Srelease: LRSp -> LR + Sp kdiss
Sdephos:  Sp -> S kdephos
```

pSTAT now reaches a steady state set by the balance of activation and dephosphorylation.



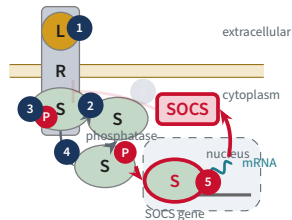
Step 5: pSTAT induces SOCS expression

```
# pSTAT binds the SOCS promoter
Gbind: Sp + G <-> GSp kGon, kGoff
# Bound promoter is transcribed
Transcribe: GSp -> GSp + mSOCS ktx
# mRNA degradation
mRNAdeg: mSOCS -> 0 kmdeg
# Translation
Translate: mSOCS -> mSOCS + SOCS kt1
# SOCS protein degradation
SOCSdeg: SOCS -> 0 kSOCSdeg
```



Step 5: pSTAT induces SOCS expression

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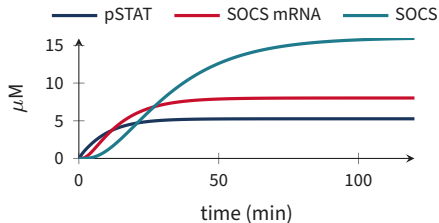
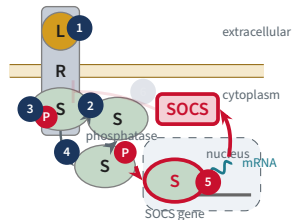
- **Synthesis:** a species on both sides (GSp, mSOCS) is a template and is not consumed.
- **Degradation:** 0 means nothing; the molecule is removed.

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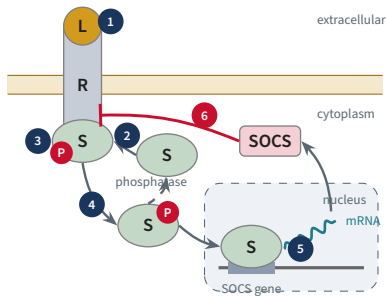
SOCS accumulates, but it has no target yet, so pSTAT is unaffected.



Step 6: SOCS binds the receptor and blocks STAT recruitment



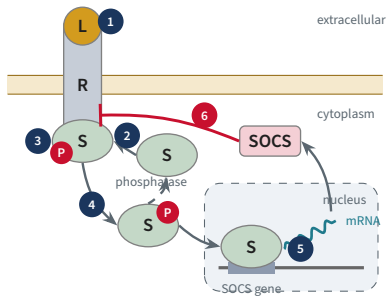
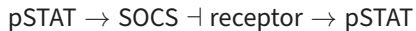
SOCS competes with STAT for the active receptor. LR·SOCS cannot recruit STAT, so pSTAT production drops.



Step 6: SOCS binds the receptor and blocks STAT recruitment



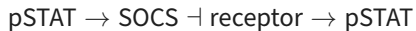
SOCS competes with STAT for the active receptor. LR·SOCS cannot recruit STAT, so pSTAT production drops. The loop is closed:



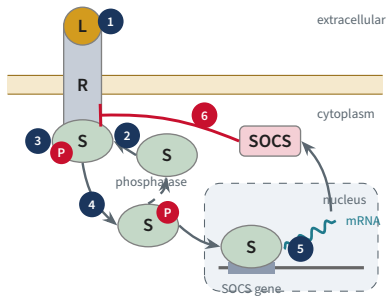
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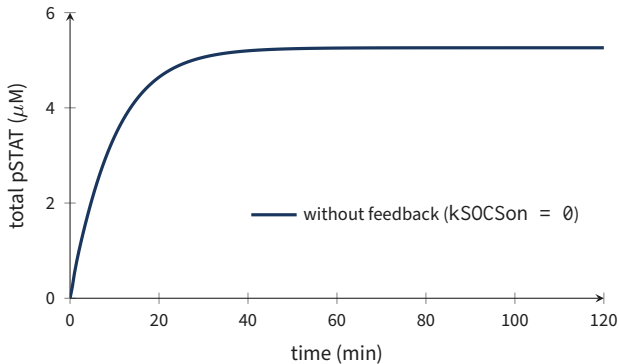
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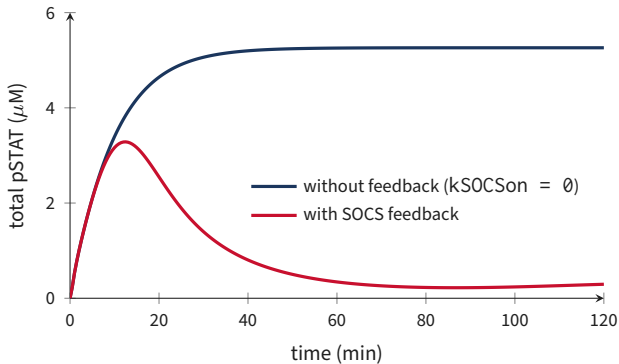
```
kSOCSon      1      # (1/uM/min)
kSOCSoFF     0.01   # (1/min)
LRSOCS
SOCSbind: LR + SOCS <=> LRSOCS kSOCSon, kSOCSoFF
```



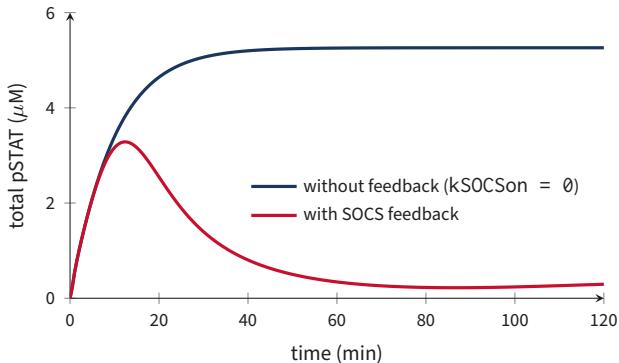
Negative feedback turns a sustained signal into a pulse



Negative feedback turns a sustained signal into a pulse

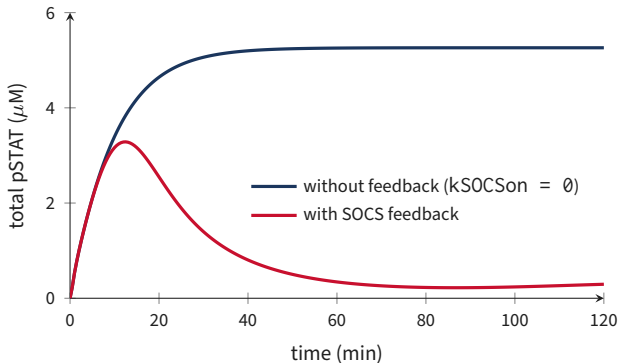


Negative feedback turns a sustained signal into a pulse



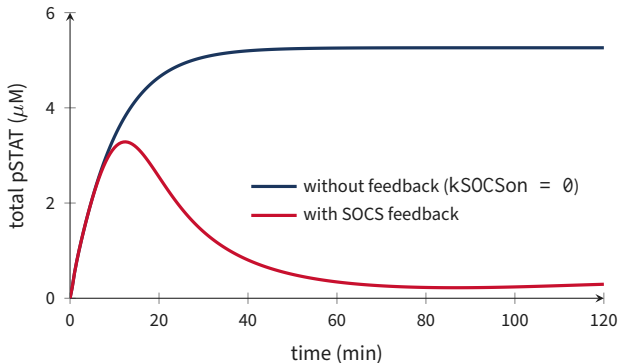
- Early response is identical: SOCS has not been made yet.

Negative feedback turns a sustained signal into a pulse



- ▶ Early response is identical: SOCS has not been made yet.
- ▶ As SOCS accumulates it sequesters receptors, and pSTAT falls to a low level despite continued ligand.

Negative feedback turns a sustained signal into a pulse



- ▶ Early response is identical: SOCS has not been made yet.
- ▶ As SOCS accumulates it sequesters receptors, and pSTAT falls to a low level despite continued ligand.
- ▶ The delay comes from transcription and translation.

Both curves come from `stat_step6_feedback.bng1`, which runs the model twice using `setParameter`.

The complete network

	reaction	rate
1	$L + R \rightleftharpoons LR$	k_{on}, k_{off}
2	$LR + S \rightleftharpoons LRS$	k_{Son}, k_{Soff}
3	$LRS \rightarrow LRS_p$	k_{act}
4	$LRS_p \rightarrow LR + S_p$	k_{diss}
	$S_p \rightarrow S$	k_{dephos}
5	$S_p + G \rightleftharpoons GS_p$	k_{Gon}, k_{Goff}
	$GS_p \rightarrow GS_p + mSOCS$	k_{tx}
	$mSOCS \rightarrow 0$	k_{mdeg}
	$mSOCS \rightarrow mSOCS + SOCS$	k_{tl}
	$SOCS \rightarrow 0$	$k_{SOCSdeg}$
6	$LR + SOCS \rightleftharpoons LR \cdot SOCS$	$k_{SOCSon}, k_{SOCSoff}$

The complete network

	reaction	rate
1	$L + R \rightleftharpoons LR$	k_{on}, k_{off}
2	$LR + S \rightleftharpoons LRS$	k_{Son}, k_{Soff}
3	$LRS \rightarrow LRS_p$	k_{act}
4	$LRS_p \rightarrow LR + S_p$	k_{diss}
	$S_p \rightarrow S$	k_{dephos}
5	$S_p + G \rightleftharpoons GS_p$	k_{Gon}, k_{Goff}
	$GS_p \rightarrow GS_p + mSOCS$	k_{tx}
	$mSOCS \rightarrow 0$	k_{mdeg}
	$mSOCS \rightarrow mSOCS + SOCS$	k_{tl}
	$SOCS \rightarrow 0$	$k_{SOCSdeg}$
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- 11 lines of BNGL generate **15 reactions** (4 are reversible) among **12 species**.

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2	$LR + S \rightleftharpoons LRS$	k_{Son}, k_{Soff}
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4	$LRS_p \rightarrow LR + S_p$	k_{diss}
	$S_p \rightarrow S$	k_{dephos}
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Part 5

Looking ahead

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Module 2: represent molecules with internal structure (binding sites and states) and write rules that generate the network automatically.

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We return to stochastic simulation in a later module.

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No local setup yet? Paste any model from [models/](#) into [BNG Playground](#) ([ruleworld.github.io/bngplayground](#)) and run it in your browser.

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- ▶ Building the cytokine–STAT–SOCS circuit one step at a time shows how each mechanism contributes; negative feedback produces an adaptive pulse of pSTAT.
- ▶ Writing every complex as a separate species quickly becomes unmanageable. This motivates rule-based modeling, the subject of Module 2.